Discrete Mathematics

Monsoon 2025

Welcome!

Evaluation

1.4 assignments: 12 marks

Bonus: If you write & submit an assignment in Latex then you will get 20% extra of your score in that assignment as bonus.

Max Bonus : 2.4 marks

Assignments' submission policy:

a) Every assignment must be submitted within 10 days.

b) Graded assignments will be returned within two weeks.

c) Assignment submission deadline won't be extended under any circumstance.

d) Assignments must be submitted on Google classroom only.

Evaluation (cont.)

2. 4 surprise quizzes: 20 marks

a) Will be conducted during tutorials.

b) The objective will be to test understanding of students.

3. Mid-sem exam : 30 marks

4. End-sem exam : 38 marks

5. Bonus : At most 3 marks

Will be given for class participation:

a) For asking relevant questions.

b) For answering questions during lectures.

Plagarism:

- Refer to the plagarism policy of IIITD.

Any deviation from it would be treated very strictly.

- The main objective of giving assignments is to give you all an opportunity to solve various problems related to the concepts taught in the lectures.
- You are strongly encouraged to solve assignments on your own & strongly discouraged to use any AI based tool like chatgpt.

Course staff

- Instructor: Nikhil Gupta
- Teaching assistants: There are total 11 TAs. The head TAs are Arpit Kumar & Debashish Sahoo.

Course passing policy:

To not recieve an F grade, you must score at least 30% cumulatively in quizzes, mid-sem & end-sem.

- Tentative course outline

1) Propositional & first-order logic : 4 lectures
2) Proofs : 6-7 lectures
3) Set theory : 8-9 lectures
4) Combinatorics : 3-4 lectures
5) Basic graph theory : 3-4 lectures

- Textbooks

a) Mathematics for computer science by Eric lehman et al (free PDF available). - Will follow it closely

b) Discrete Mathematics & its applications, 8th edition, by Kenneth H. Rosen.

Other important details

a) Class time : 9:30 am – 11 am on Mondays & Wednesdays at B003 (R&D Building)

b) Tutorials : 2:00 pm – 3:00 pm at C01 & C11.

c) Office hour : 2 – 3 pm on Tuesdays at B-511 (R&D building).

d) Google classroom class code: **qzliysdd**

Motivation:

1) Develops critical thinking, needed for problem solving.

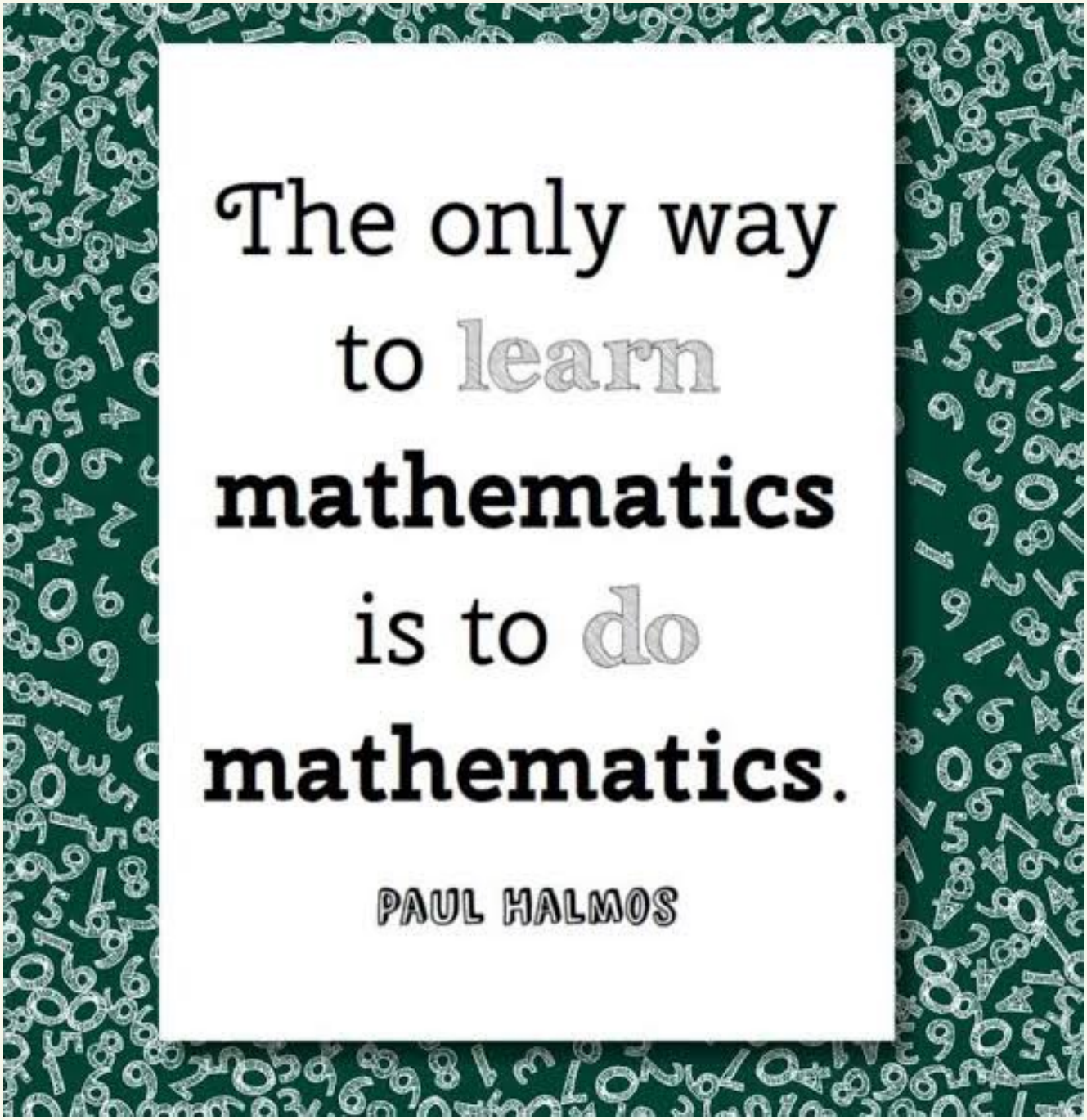
2) Trains students in proving things, which is extremely important in maths & computer science.

3) Required for various core subjects of computer science like algorithms, theory of computation, data structures, databases, etc.

4) Prepares students for research.

5) It is a gateway to advanced mathematics.

Some famous quotes



MATHEMATICS
is not about
numbers, equations,
computations, or
algorithms:
it is about
UNDERSTANDING.

William Paul Thurston

11th August Lecture 1

Mathematical logic

- Propositional logic
- First-order logic

Definition [Proposition]: A proposition is a declarative statement which is either true or false but not both.

Examples:

a) The Sun rises in the East.

b) $1+1=4$

Examples of statements which are not propositions.

a) How are doing?

b) Happy Diwali!

c) $x+1=3$, where x is a variable. → It is a declaration.

This can be turned into a proposition by substituting x with a number.

For example, for $x=2$, it is true, but for $x=3$, it is false.

Propositions are building blocks of logic.

There are declarative statements which can be simultaneously true & false. Such statements are called paradoxes.

H.W. Read about Russell's paradox.

It is not always easy to determine the truth value of a proposition.

Example: For every natural number < 40, $n^2 + n + 41$ is a prime number. True!

Natural numbers: $0, 1, 2, 3, \ldots$

Definitions [Prime number]:
A natural number a is called prime if $a > 1$ and a is divisible only by 1 & itself.

Remark: If 1 is prime then the 'Fundamental theorem of arithmetic' fails.

Notation [Univariate polynomial]:
Let x be a variable. Then, a univariate polynomial in x looks as

$f = a_d x^d + a_{d-1} x^{d-1} + \dots + a_1 x + a_0$

E.g. $5x^2 + 6x + 3$

where $x^d, \dots, x$ are called as monomials & $a_d, \dots, a_0$ are called coefficients. We assume that these coefficients are natural numbers.

If $a_d \neq 0$ then d is called the degree of f. The degree of f can never be negative.

If $a_d = 0, \dots, a_1 = 0$ then f is called a constant polynomial.

For a natural number n,

$$f(n) :\equiv a_d n^d + \cdots + a_1 n + a_0$$

$:\equiv$ → symbol used for defining.

Theorem 1: Let $f = a_d x^d + \cdots + a_1 x + a_0$ be an arbitrary polynomial, where $a_d, \ldots, a_0$ are natural numbers. Then, not for every natural number n, $f(n)$ is a prime number.

13th August Lecture-2

Recap

1) Propositions, examples
2) Natural numbers, prime numbers
3) Univariate polynomials, constant & non-constant polynomials
4) Paradoxes.

Theorem 1: Let $f = a_d x^d + \cdots + a_1 x + a_0$ be a non-constant polynomial and $a_0, \ldots, a_d$ be natural numbers. Then, **not** for every natural number n, $f(n)$ is a prime number.

Proof: Given $f = a_d x^d + \cdots + a_1 x + a_0$

To show: Not for every natural number n, $f(n)$ is prime.

Put $n = a_0$

$$f = a_d a_0^d + \cdots + a_1 a_0 + a_0 \neq 0$$

$$= a_0 \underbrace{(a_d a_0^{d-1} + \cdots + a_1 + 1)}$$

$a_0 \stackrel{?}{=} 1$.

Problem 1

$\Downarrow$

Put $n = 0$

$f(0) = 1$

$a_0 \stackrel{?}{=} 0$.

Problem 2.

$\Downarrow$

$f = a_d x^d + \cdots + a_1 x$

$= x(a_d x^{d-1} + \cdots + a_1)$

Put $n = 6$

$f(6) = 6 \underbrace{(a_d 6^{d-1} + \cdots + a_1)}$

$\neq 1$.

Definition [Conjecture]: It is a proposition which is neither known to be true nor false.

Examples:

a) Goldbach's conjecture:

Every even natural number greater than 2 is the sum of two prime numbers.

E.g: $6 = 3+3$, $4 = 2+2$

$8 = 5+3$ ...

Stated in 1742. Wide open.

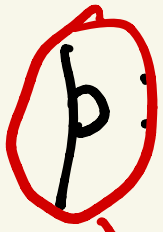
Researchers have verified that every even number less than equal to 10^{18} can be written as the sum of

2 prime numbers.

2) The twin prime conjecture:
A prime number p is called as a twin prime if either $p=2$ or there exists a prime number q such that $p=q+2$.
E.g. 5 , $p=5, q=3$.
There are infinitely many twin primes.
First stated in 1849 & it is open since then.

1.1 Operations on propositions

We will denote propositions by variables. For example, $1+2=4$ is proposition. We can write it as $p: 1+2=4$



p ↘ propositional variable.

One benefit of representing propositions with variables is that we can apply many operations on them & get "bigger" propositions, which are called <u>compound propositions</u>.

Definition [Propositional logic]: It is a branch of mathematics that deals with propositions & operations on them.

We now list important operations on propositions.

1) The negation operation: Let p be a proposition. Then, the negation of p, denoted as $\neg p$, is the new proposition which is true whenever p is false & it is false whenever p is true. This is captured by the following

table, called the truth table of negation.

p	$\neg p$
True	False
False	True

- Negation is a unary operation

Example: p: It is raining.

$\neg p$: It is not raining.

2. The conjunction operation (or the And operation): Let p & q be propositions. Then, the conjunction of p, q, denoted $p \wedge q$, is the proposition which is true whenever both p & q are true, otherwise it is

false. Its truth table is as follows.

T: True
F: False

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

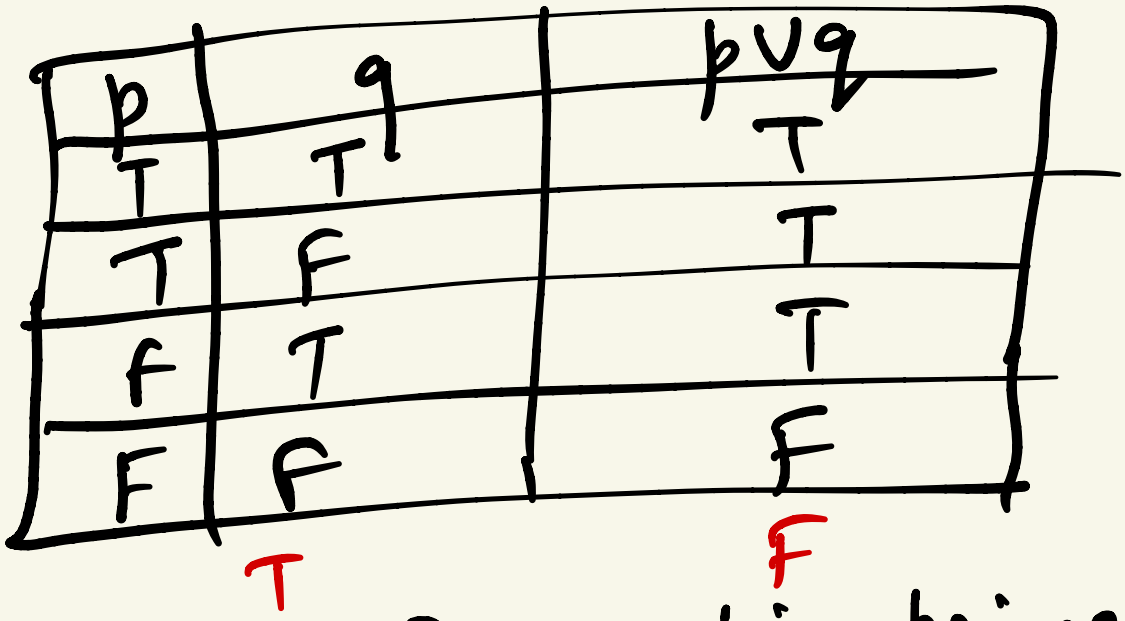
T F

Let p: 1+2=3, q: 1 is a prime.

$p \wedge q$ is false.

3. The disjunction operator (or the OR operator): Let p, q be propositions. Then, the disjunction of p, q, denoted $p \vee q$, is the proposition which is true if at least one of p, q is true, otherwise it is false.

It truth table is

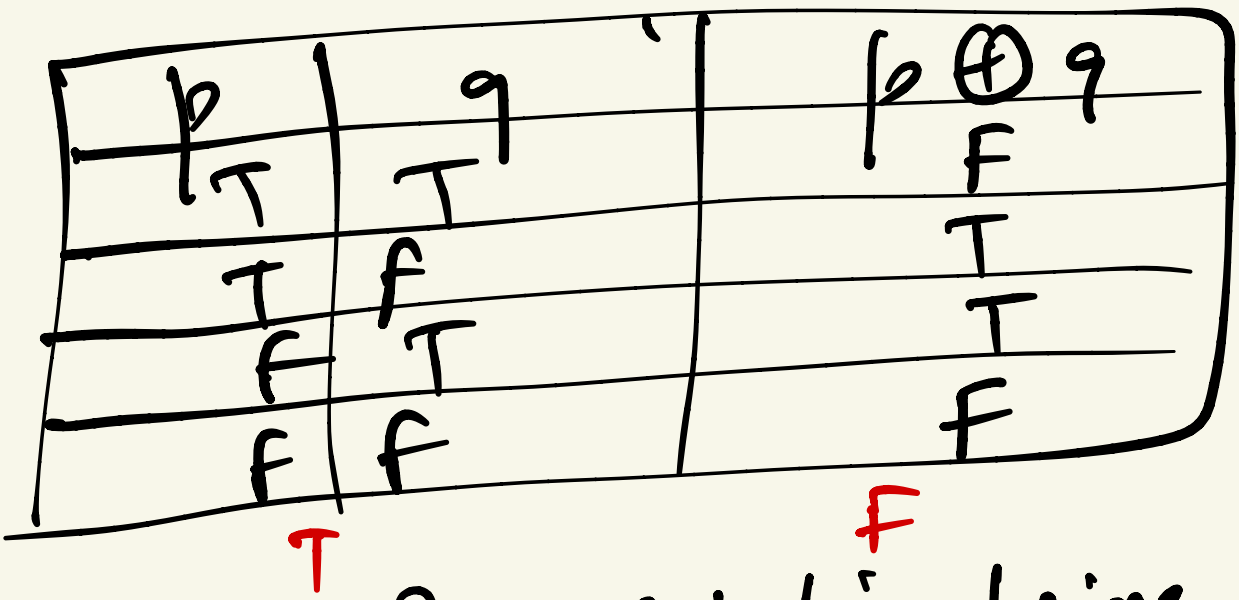


Eg: p: $1+2=3$, q: 1 is prime.

$p \vee q$ is true.

4. The Exclusive-OR operator:

Let p, q be propositions. Then, the exclusive-or of p, q, also called XOR of p, q, denote $p \oplus q$, is the proposition that is true whenever

exactly one of p, q is true & the other is false, otherwise $p \oplus q$ is false. Its truth table is

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

T F

E.g. p: $1+2=3$, q: 1 is prime

$p \oplus q$ is True.

5. The implication operation (or "if, then" operation or conditional operation): Let p, q be propositions. Then, the

Statement if p then q, denoted $p \Rightarrow q$, is the proposition, which is false whenever p is true but q is false, otherwise $p \Rightarrow q$ is true. Its truth table is as follow.

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Examples: a) p: Goldbach's conjecture is true. Not known to be T or F

b) q: $x^2 \geq 0$ for every integer x. T

Here, $p \Rightarrow q$ is true.

Thumb rule: Whenever you see a logical operation, use its truth table & not the intution.

Some important terminologies.

a) In $p \Rightarrow q$, p is called the hypothesis (or premise) & q is called the conclusion (or consequence).

b) $p \Rightarrow q$ is also interpreted as follows:

i) p is sufficient for q,
ii) q is necessary for p.

c) **Converse**: The proposition $q \Rightarrow p$ is called the converse of $p \Rightarrow q$.

d) **Contrapositive**: The proposition $(\neg q) \Rightarrow (\neg p)$ is called the contrapositive of $p \Rightarrow q$.

e) **Inverse**: The proposition $(\neg p) \Rightarrow (\neg q)$ is called the inverse of $p \Rightarrow q$.

Homework: a) Show that the proposition & its contrapositive have the same truth tables.

b) Show that the truth table of the converse & the inverse of $p \Rightarrow q$ are the same.

p	q	$\neg p$	$\neg q$	$q \Rightarrow p$	$\neg p \Rightarrow \neg q$
T	T	F	F	T	T
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	T	T	T

$p \Rightarrow q$

$q \Rightarrow p$

$\neg p \Rightarrow \neg q$

20th August

Lecture 3

Recap:

1) Propositions, paradoxes & conjectures.

2) Operations on propositions:

unary ← $\neg, \wedge, \vee, \oplus, \Rightarrow$ → binary

3) Converse, inverse & contrapositive of an implication.

6) Biconditional (or If and Only If)

Let p, q be propositions. Then the statement "p if and only if q" denoted $p \Longleftrightarrow q$, is the

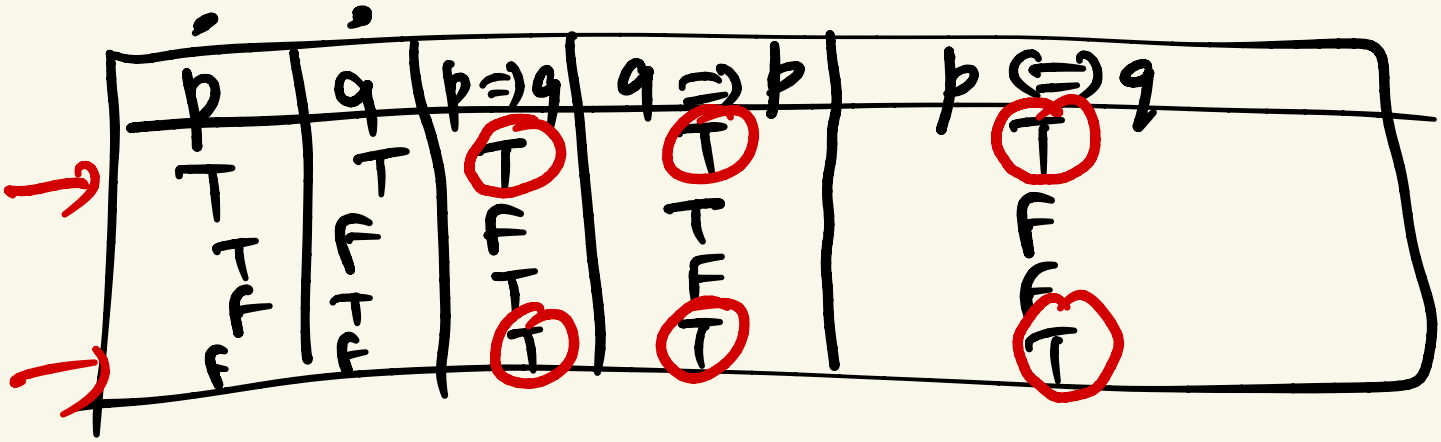
compound proposition which is true when both p & q have the same truth value, otherwise $p \Leftrightarrow q$ is false. Its truth table is

	p	q	$p \Leftrightarrow q$
→	T	T	T
	T	F	F
	F	T	F
→	F	F	T

$p \Leftrightarrow q$ is true whenever both

$p \Rightarrow q$ & $q \Rightarrow p$ are true.

logical and



We have shown that $p \Longleftrightarrow q$ and $(p \Rightarrow q) \wedge (q \Rightarrow p)$ have the same truth tables.

- $p \Longleftrightarrow q$ is also called "p is necessary and sufficient for q."

Notations:

- The symbols used to represent logical operations are called operators.
- The propositional variables

to which an operator is applied are called operands.

Definition [Propositional formula]: A proposition containing propositional variables & operators is called a propositional formula.

for example, $p \vee q$.

- We can also include parantheses in the list of operators. A paranthesis is used for grouping.

Question: Suppose we have a

compound proposition containing more than one operators. How do we evaluate it?

E.g. $p \wedge q \vee r \Rightarrow s$

To resolve this, we have the notion of precedence of logical operators. It is similar to the BODMAS rule.

Operators	Precedence
$( )$	1
$\neg$	2
$\wedge$	3
$\vee$	4
$\Rightarrow$	5
$\Leftrightarrow$	6

Example: Let p, q, r be propositions.

Write the truth table of

$$u := \boxed{\neg(p \Rightarrow q)} \vee \boxed{r \wedge \neg p}$$

p	q	r	$(p \Rightarrow q)$	$\neg(p \Rightarrow q)$	$\neg p$	$r \wedge \neg p$	u
T	T	T	T	F	F	F	F
T	T	F	T	F	F	F	F
T	F	T	F	T	F	F	T
T	F	F	F	T	F	F	T
F	T	T	T	F	T	T	T
F	T	F	T	F	T	F	F
F	F	T	T	F	T	T	T
F	F	F	T	F	T	F	F

Homework: Write the truth table of $u := \neg p \wedge q \Longleftrightarrow \neg r \Rightarrow p \vee r$.

1.2 The algebra of propositions.

Definition [Equivalent propositions]:

We say that two propositions

p and q are equivalent if and only if they have the same truth tables. In other words, the following two conditions are satisfied:

- If p is T then q is T
- If p is F then q is F.

E.g. $p \Rightarrow q$ & $\neg q \Rightarrow \neg p$ are equivalent.

Motivating question: We can come up with an endless list of propositions and each proposition denotes a logical operation. It is natural to as if there

exists a small list of logical operators such that every compound proposition has an equivalent proposition containing operators only from the small list.

Notation: Suppose p, q are equivalent propositions. Then, they are denoted as $p \equiv q$.

We will show that this list of $\neg, \vee, \wedge$ "capture" every compound proposition.

Definition [Disjunctive form]: A proposition is said to be in the disjunctive form if it is an OR of AND-terms, where every AND-term is the AND of propositional variables with or without negation.

E.g. $(p \wedge q) \vee (q \wedge \neg r)$.

AND terms

Definition [Disjunctive normal form] A disjunctive form is said to be a disjunctive normal form (DNF) if every AND-term contains

every propositional variable with or without negation.

E.g. $(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r)$ is a DNF

but

$(p \wedge q) \vee (p \wedge \neg r)$ is not a DNF.

- $p \Rightarrow q$ is not in disjunctive form.
- **Obtaining a DNF for a proposition:** Let u be a compound proposition. We now present a procedure to obtain a

DNF that is equivalent to u, from the truth table of u.

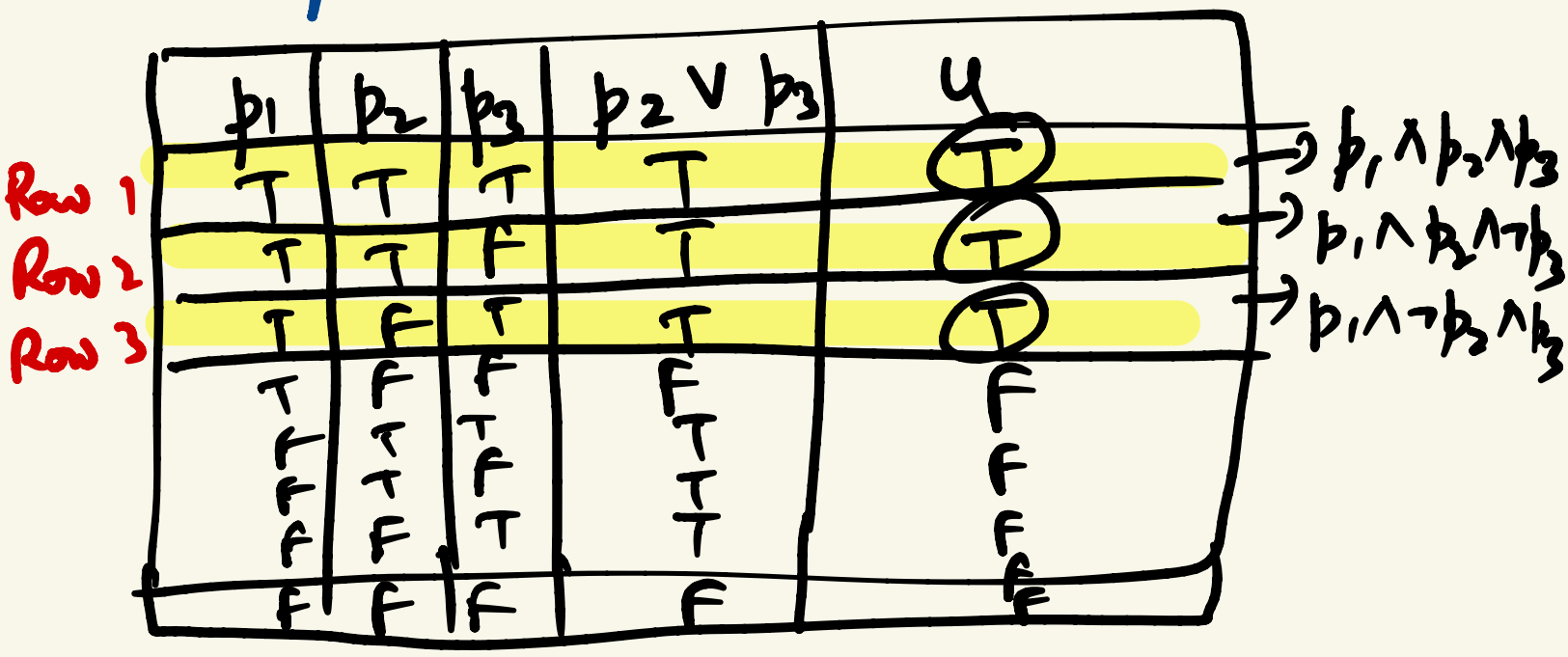
Suppose u depends on propositional variables $p_1, \ldots, p_n$ for some natural number n.

1. Write the truth table of u.
2. Mark all the rows in this truth table whose final value is true.
3. Obtain AND-terms corresponding to the rows marked above as follows: Pick any marked row. For every $1 \leq i \leq n$, if the propositional

variable p_i in this row is T, then add p_i to the corresponding AND-term, otherwise add $\neg p_i$

4. The final DNF is the OR of the AND-terms of all the marked rows.

Example: Let $u := p_1 \wedge (p_2 \vee p_3)$.



AND-term for row 1:
$p_1 \wedge p_2 \wedge p_3$

AND-term for row 2: $p_1 \wedge p_2 \wedge \neg p_3$

AND-term for row 3: $p_1 \wedge \neg p_2 \wedge p_3$

The final DNF:

$v := (p_1 \wedge p_2 \wedge p_3) \vee (p_1 \wedge p_2 \wedge \neg p_3) \vee (p_1 \wedge \neg p_2 \wedge p_3)$.

To prove: $u \equiv v$.] Homework.

23rd August Lecture-4

Recap

a) Propositions, conjectures, paradoxes
b) operations on propositions:
$\neg, \vee, \wedge, \oplus, \Rightarrow, \Leftrightarrow$
c) Equivalent propositions
d) Disjunctive forms & DNFs.

Let u be the given proposition & v be the DNF obtained from the procedure mentioned above. We want to prove that $u \equiv v$. One immediate approach is to verify that

they have the same truth tables. But it becomes inefficient if we have lot of p_is.

Another approch to verify the equivalence is as follows:

Let us pick (arbitrarily) an AND term from the Step 3 of the procedure given above.

We know that this comes from a row of the truth table of u where u is T.

Suppose the values of $p_1, \ldots, p_n$ in this row are $a_1, \ldots, a_n$, where every a_i is either T or F.

Then, this AND-term looks like

$w := \hat{p}_1 \wedge \hat{p}_2 \wedge \cdots \wedge \hat{p}_n$,

where $\hat{p}_i$ is equal to p_i if a_i is T, otherwise $\hat{p}_i = \neg p_i$.

E.g. We have p_1, p_2. $a_1 = T, a_2 = F$. $w = p_1 \wedge \neg p_2$

Now, lets take any other row of the truth table of α & suppose $p_1, \ldots, p_n$ have values $b_1, \ldots, b_n$ in this row.

Observe that there exists an i such that $a_i \neq b_i$.

Now, as w is an AND-term, note that on sustituting $b_1, \ldots, b_n$ in w, we get F.

We want to show that

$u \equiv v$ and V is the OR of many AND-terms like w.
Take a row from the truth table of u which is T. Look at its corresponding AND term. It will always be true & as v is OR of such AND-terms, we get that v is true for this row. This proves that whenever u is true v is also true.
We have to now show that whenever u is false, v is

also false.

Let us arbitrarily pick a row where u is false. As discussed above, every AND-term of v becomes false for this row & as v is OR of AND-terms we get that v is false for this row.

This proves $u \equiv v$.

Remark: This proof holds for every compound proposition u.

Definition [Conjunctive form]: A proposition is said to be in the conjunctive form if it is an AND of the OR-terms, where every OR-term is OR of propositional variables with or without negation.

E.g. $(p_1 \vee p_2) \wedge (p_2 \vee \neg p_3)$.

Definition [Conjunctive normal form (CNf)]: A proposition in the conjunctive form is in CNf if every OR term contains every variable with or without negation.

E.g: $(p_1 \vee p_2 \vee p_3) \wedge (\neg p_1 \vee p_2 \vee p_3)$

Obtaining a equivalent CNF of a proposition from its truth table: Let u be a propositional formula defined using variables $p_1, \ldots, p_n$.

a) Write the truth table of u.

b) Mark the rows of the truth table of u where the final value is F.

c) We obtain OR-terms for marked rows as follows: Pick a marked row arbitrarily. For every $1 \leq i \leq n$, if the value of p_i in this row is T, then put $\neg p_i$ in the OR-term, otherwise put p_i in the OR-term.

called v_G

4) The final CNF is the AND of all the OR-terms

obtained above.

Homework: Prove that $u \equiv v$, where v is the final CNF obtained above.

This question will be in the assignment.

HW: Obtain the CNF equivalent to $p_1 \wedge (p_2 \vee p_3)$

Theorem 1: Let u be an arbitrary proposition. Then, there exists a CNF v & a DNF w s.t. $u \equiv v \equiv w$.

1.2.2 Proving equivalence of propositions (without using truth tables):

It is a very natural question to check if two propositions are equivalent. We saw that one way of doing it is to check if they have the same

truth tables. But the problem with this approach is that if we have n basic variables then there are 2^n many rows in the truth table & if n is large, this approach becomes inefficient!

Now we present an alternate approach for testing equivalence of propositions. It is based on the algebra of propositional logic.

- The first step in this approach is to get rid of all operators other than

$\neg, \wedge, \vee$ (called basic operators). This means that we replace propositions using non-basic operators with equivalent propositions using only basic operators.

Non-basic operators	Equivalent propositions containing only $\neg, \wedge, \vee$
$p \Rightarrow q$	$\neg p \vee q$
$p \oplus q$	$(p \wedge \neg q) \vee (\neg p \wedge q)$
$p \Leftrightarrow q$	$(p \wedge q) \vee (\neg p \wedge \neg q)$

Table I

Homework: Prove these equivalences.

Rules of the algebra of propositional logic: Let p, q, r be propositional variables & T, F denote true & false. Then, the

rules are stated below:

1) Rules for AND:

a) Commutativity of AND: $p \wedge q \equiv q \wedge p$

b) Associativity of AND:

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r.$$

c) Identity for AND: $p \wedge T \equiv p$

d) Falsity for AND: $p \wedge F \equiv F$

e) Idempotence for AND: $p \wedge p \equiv p.$

f) Contradiction for AND: $p \wedge \neg p \equiv F$

2) Rules for OR:

a) Commutativity of OR: $p \vee q \equiv q \vee p$

b) Associativity of OR:

$$p \vee (q \vee r) \equiv (p \vee q) \vee r.$$

c) Identity for OR: $p \vee F \equiv p$

d) Truth for OR: $p \vee T \equiv T$

e) Idempotence for OR: $p \vee p \equiv p$

f) Validity for OR: $p \vee \neg p \equiv T$

3) Other rules:

a) Double negation: $\neg\neg p \equiv p$

b) Distributivity of AND over OR:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

c) Distributivity of OR over AND:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

d) DeMorgan's law:

i) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

ii) $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Homework: Verify all logical

equivalences given in these rules.

The rules stated above are all that are needed to convert any propositional formula to its equivalent CNF & DNF without using the truth table We show how to convert a propositional formula into an equivalent CNF using the algebra of propositional logic in the following steps:

1) First, we eliminate all occurances of $\oplus$, $\Rightarrow$, $\Leftrightarrow$ using Table 1 given above.

2) We move all the negations inwards such that they appear only before propositional variables, using double negation & DeMorgan's law.

3) Then, we use other laws of the algebra to obtain an equivalent proposition in conjunctive form.

4) Convert every OR-term in the conjunctive form above to CNFs

5) Eliminate all redundant OR-terms.

Remark: By replacing OR-terms with AND-terms the conjunctive form with disjunctive form, and CNF with DNF in the above procedure, we obtain a DNF equivalent to the given proposition.